THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** hypotonic to the cell	Water entered.
7	**(d)** accumulation of food	Phloem blocked.
8	**(c)** Fungi	Hyphae and chitin.
9	**(d)** 14 March	Mid cycle.
10	**(b)** 4000 J	400 W x 10 s.
11	**(c)** synthesis of ATP	Mitochondria do ATP.
12	**(d)** glandular epithelium	Secretion.
13	**(b)** They are in the same class and phylum	Higher ranks shared.
14	**(b)** the one with an eye grows	Bud needed.
15	**(c)** algal bloom and fish kill	Eutrophication.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	SA/V = 6 and 2 (1); Larger cell exchanges less efficiently, so it divides (1).	1+1
17	Secondary xylem accumulates every year (1); Lignified walls give strength (1).	1+1
18	New evidence from molecular studies (1); Discovery of new organisms (1).	1+1
19	Muscle 46, sperm 23 (1); Egg 23, zygote 46 (1).	1+1
20	Absorbed = 0.7 x 342 (1); = 239 W/m squared (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	0.5 per cent: gains mass (1); 5 per cent: loses mass (1); 1 per cent is isotonic with the potato cell sap (1).	1+1+1
22	Potato turns blue-black as it stores starch (1); Carrot gives less blue-black (1); Parenchyma stores food (1).	1+1+1
23	Feathers? yes: pigeon (1); No: has hair? yes: bat (1); No: gills with scales: shark; moist skin: frog; dry scaly skin: snake (1).	1+1+1
24	Ovulation about 14 days before the next period: day 16 (1); Fertile period: days 13 to 17 (1); Sperms survive 3 days before (1).	1+1+1
	OR: Annual emission = 250 x 12 x 0.8 = 2400 kg (1); Trees = 2400/22 (1); = 109 trees (1).	1+1+1
25	Break cells gently (1); Centrifuge at low speed to remove nuclei (1); Higher speed to get mitochondria (1).	1+1+1
	OR: Constant energy demand (1); Continuous contraction (1); Supplies ATP (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	Cut equal potato cubes, weigh (1); Place in solutions for same time (1); Reweigh and calculate the percentage change (1); Plot the graph of percentage change vs concentration (1); Intercept on the x-axis gives isotonic concentration (1).	1+1+1+1+1
	OR: Place a twig in red ink (1); Section shows red xylem (1); Girdle a branch (1); Bulge above the ring (1); Conclusions about xylem and phloem (1).	1+1+1+1+1
27	Cell type: prokaryote - bacterium (1); Eukaryote unicellular: Amoeba (no wall), yeast (wall) (1); Multicellular plants: vascular? no: moss; yes: seeds? no: fern (1); Seeds: cones - pine; flowers - rose (dicot), wheat (monocot) (1); Characters: cell wall, vascular tissue, seeds, cotyledon (1).	1+1+1+1+1
	OR: Tube 1: moist cotton, room temperature (1); Tube 2: dry cotton (1); Tube 3: seeds submerged in boiled water with oil (1); Tube 4: moist cotton in a fridge (1); Only tube 1 germinates; conclusion (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** cell wall	Rigid wall.
2	**(d)** sensory neuron, relay neuron and motor neuron	Reflex arc.
3	**(d)** animal	Animalia.
4	**(a)** wings or hairs	Light structures.
5	**(d)** the ground is heated by sunlight and warms the air	Surface heating.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** lignin	Lignified walls.
8	**(a)** dolphins have lungs, hair and mammary glands	Mammalia vs Pisces.
9	**(a)** oviduct	Tube.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** hypertonic to A and hypotonic to B	Different internal concentrations.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
15	**(a)** increase	More absorption.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	It protects the cell and controls entry and exit of substances (1); It allows only some substances to pass (1).	1+1
17	Xylem upward only, phloem both directions (1); Xylem passive; phloem needs energy (1).	1+1
18	Homeothermic: birds and mammals (1); Poikilothermic: fish, amphibia, reptiles (1).	1+1
19	Pollen grains carry male gametes (1); Ovule has the egg; becomes seed after fertilisation (1).	1+1
20	Long coastline and monsoon dependence (1); Large population in agriculture (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Moss: gametophyte dominant (1); Fern: sporophyte dominant (1); Both need water for fertilisation (1).	1+1+1
22	Seeds from sexual reproduction give variation (1); Cutting is a clone (1); Mitosis vs meiosis and fertilisation (1).	1+1+1
23	Dry air has little water vapour to trap heat (1); Quick cooling at night (1); Bare sand heats up fast (1).	1+1+1
24	Fill the bag with starch and glucose (1); Place in water (1); Glucose passes out; starch stays (1).	1+1+1
	OR: Vascular cambium: secondary xylem and phloem (1); Cork cambium: cork (1); Together form secondary growth (1).	1+1+1
25	Include unrelated organisms (1); Based on a few characters (1); Different evolutionary origins (1).	1+1+1
	OR: Unisexual flowers (1); Anthers and stigma mature at different times (1); Different positions (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Absorbed = 0.7 x 342 = 239 W/m squared (1); At equilibrium the outgoing is 239 W/m squared (1); Albedo 0.28: absorbed 246 W/m squared, warming (1); More greenhouse gases reduce the outgoing radiation (1); Surface warms until balance is restored (1).	1+1+1+1+1
	OR: Stem cells unspecialised (1); Differentiation into RBC, nerve, muscle (1); Plant meristems (1); Uses: bone marrow transplants (1); Ethical issues: source of embryonic cells (1).	1+1+1+1+1
27	Xerophytes: thick cuticle, sunken stomata (1); Hydrophytes: aerenchyma (1); Mesophytes: normal tissues (1); Root tissues for water (1); Support tissues (1).	1+1+1+1+1
	OR: Roots to absorb water (1); Vascular tissue for transport (1); Seeds protect the embryo (1); Flowers for pollination (1); Fruits for dispersal (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** bacterial	Prokaryote.
2	**(d)** xylem is intact	Water still moves.
3	**(a)** Gymnosperms	Cones with naked seeds.
4	**(d)** 13 to 15 March	Around ovulation.
5	**(a)** water filtration and flood control	Natural sponge.
6	**(c)** large central vacuole	Takes up space.
7	**(c)** a single nucleus and spindle shape	Unstriated.
8	**(d)** belong to the same genus but different species	Same genus Canis.
9	**(a)** cambia of stock and scion are in contact	Tissue contact.
10	**(c)** absorption of UV by ozone	Ozone heating.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(b)** dendrite to axon	Direction.
13	**(c)** Plantae	Plant cell.
14	**(b)** hooks or fleshy fruit	Burrs.
15	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Protoplasm separates from the cell wall (1); Gap between wall and membrane (1).	1+1
17	No growth in length or girth (1); No new organs would develop (1).	1+1
18	Characteristic of Chordata (1); Used to identify the groups (1).	1+1
19	IVF (test-tube baby) (1); Artificial insemination (1).	1+1
20	UV has the energy to break O2 and O3 bonds (1); Visible light has less energy (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	30 units are reflected and 70 are absorbed (1); The Earth radiates back 70 units in equilibrium (1); Greenhouse gases delay the escape of heat (1).	1+1+1
22	Root tip: mitosis, growth (1); Testis: meiosis for gametes (1); Chromosome numbers (1).	1+1+1
23	Dicot stem (1); Vascular bundles in a ring (1); Open bundles with cambium (1).	1+1+1
24	Fungi (1); Heterotrophic by absorption (1); Eukaryotic with a cell wall (unicellular like yeast) (1).	1+1+1
	OR: Frog: thousands of eggs, no care (1); Hen: few eggs, some care (1); Human: one baby, extensive care (1).	1+1+1
25	Saving = 20 W x 100 = 2 kW (1); Energy = 2 x 8 x 250 = 4000 kWh (1); CO2 = 3200 kg (1).	1+1+1
	OR: They contain chlorophyll (1); They trap sunlight (1); They make food (glucose) by photosynthesis (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Causes: blocked tubes, low sperm count (1); Hormonal problems (1); IVF steps (1); Artificial insemination (1); Social acceptance and ethics (1).	1+1+1+1+1

	OR: Less transpiration, less rainfall (1); Carbon released (1); Albedo changes (1); Biodiversity loss (1); Soil erosion and floods (1).	1+1+1+1+1
27	Prokaryotic: no true nucleus, no membrane-bound organelles (1); Eukaryotic: nucleus and organelles (1); Eukaryotic cells divide the work among organelles (1); Viruses have no cell structure (1); Viruses multiply only inside host cells (1).	1+1+1+1+1
	OR: Dicot: ring of open bundles (1); Monocot: scattered closed bundles (1); Cambium present in dicots (1); No secondary growth in monocots (1); Examples (1).	1+1+1+1+1

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